How much original mass lies near sampled vectors?

A distance-based measure of original probability mass

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1. Question and distributions

At fixed input x, let d be a positive integer giving the vector dimension and Q_x the sampler's final distribution on $\mathbb{R}^d$. Let P_0 be the original vector distribution on the same space. Independent sampler draws are $Y_i \sim Q_x$; an original-distribution vector is $X \sim P_0$. We ask: what fraction of P_0 lies near the vectors the sampler produces?

That fraction is *original probability mass*, a number between zero and one. It is different from the geometric volume of a region in $\mathbb{R}^d$. A histogram with one horizontal axis becomes meaningful for this question after we turn each vector into a scalar distance score.

2. From vectors to a distance score

Choose a measurable distance ρ on the vector representation. Its feature scaling is part of the definition; Euclidean distance is one choice. Hold out some sampler draws and use $s \geq 1$ others as a reference set $C = \{Y_1, \ldots, Y_s\}$. For a vector z and radius $r \in [0, \infty]$, define

$$D_C(z) = \min_{y \in C} \rho(z, y), \qquad R_r = \{z : D_C(z) \leq r\}. \tag{1}$$

Thus $D_C(z)$ is the distance from z to its closest sampled vector, and R_r is the union of distance-r neighborhoods around the reference vectors.

A similarity measure gives the same construction. For example, with similarity scale $h > 0$, define

$$M(z, y) = \exp\left(-\frac{\rho(z, y)^2}{2h^2}\right), \qquad \max_{y \in C} M(z, y) = \exp\left(-\frac{D_C(z)^2}{2h^2}\right). \tag{2}$$

High similarity and small distance therefore select exactly the same vectors.

3. Choose the radius from fresh sampler draws

Let $V_1, \ldots, V_m \overset{\text{iid}}{\sim} Q_x$, with $m \geq 1$, be check draws independent of C. For a desired sampler coverage $0 < \alpha < 1$, sort their scores $t_j = D_C(V_j)$ as $t_{(1)} \leq \cdots \leq t_{(m)}$, where $t_{(j)}$ is the jth smallest score, and set

$$k = \lceil (m+1)\alpha \rceil, \qquad r_\alpha = \begin{cases} t_{(k)}, & k \leq m, \\ \infty, & k = m+1. \end{cases} \tag{3}$$

Theorem 1. For a fresh, independent $Y_{\text{new}} \sim Q_x$,

$$\Pr\{Y_{\text{new}} \in R_{r_\alpha} \mid C\} \geq \alpha, \tag{4}$$

where the probability includes the random check draws used to choose r_α.

Reason. Conditional on C, the m check scores and the fresh score are exchangeable. If $k \leq m$, the fresh score falls at or below the kth check score with probability at least $k/(m+1) \geq \alpha$; ties can only increase coverage. If $k = m+1$, the radius is infinite and coverage is one.

4. Area under the original distance histogram

For the region selected by (3), define its mass under the original distribution, conditional on the realized reference and check draws:

$$\boxed{A_\alpha = P_0(R_{r_\alpha}) = \mathbb{E}_{X\sim P_0}[\mathbf{1}\{D_C(X) \le r_\alpha\}]\,.} \tag{5}$$

Here $\mathbf{1}\{\cdot\}$ is one when its condition holds and zero otherwise. If P_0 has vector density p_0, this is

$$A_\alpha = \int_{\mathbb{R}^d} \mathbf{1}\{D_C(z) \le r_\alpha\} p_0(z)\,dz = \int_{R_{r_\alpha}} p_0(z)\,dz. \tag{6}$$

Now transform an original vector into the scalar $T_C = D_C(X)$, and let $H_C(t) = P_0(T_C \le t)$ be its cumulative distribution function for fixed C. If T_C has a density, denote it by $h_C(t)$. Then the key histogram identity is

$$\boxed{A_\alpha = H_C(r_\alpha) = \int_0^{r_\alpha} h_C(t)\,dt} \tag{7}$$

when the density exists. In plain terms: build a histogram of *distance to the closest sampled vector* for vectors from the original distribution. The area to the left of r_α is the requested original probability mass. If the score has atoms, use the cumulative distribution H_C rather than the density integral.

5. Calculation from original vectors

If $X_1,\dots,X_N \overset{\text{iid}}{\sim} P_0$, with $N \ge 1$, are original-distribution vectors independent of C and the check draws, compute their distances to C and count those within the radius:

$$\widehat{A}_\alpha = \frac{1}{N}\sum_{\ell=1}^{N} \mathbf{1}\{D_C(X_\ell) \le r_\alpha\}. \tag{8}$$

Conditional on C and the check draws, this estimate has mean A_α and variance $A_\alpha(1 - A_\alpha)/N$. A histogram is a display of the same calculation; counting before binning avoids a partial-bin approximation.

6. Reading and limits

For example, $\alpha = 0.90$ and $A_\alpha = 0.35$ means a distance-defined region calibrated to cover at least 90% of future sampler draws, in the sense of (4), contains 35% of the original distribution. The guarantee in (4) averages over the random check set; a particular realized region may have different coverage.

For scalar outputs, let F_0 be the cumulative distribution function of P_0. Choosing the interval from the smallest to largest of the s reference draws gives original mass $F_0(\max_{1\le i\le s} Y_i) - F_0(\min_{1\le i\le s} Y_i)$ when P_0 is continuous. With vectors, if p_0 exists, taking minimum and maximum *density heights* $p_0(Y_i)$ instead selects a band of density values; it need not select vectors close to any sample. Likewise, a histogram of one raw feature cannot determine (5). One needs the original full vectors or their joint distribution.

The answer depends on the representation, distance, and coverage level. Finite sampler draws define an operational neighborhood; they do not identify the full support of Q_x.

READING OF THE RESULT. Calibrate a distance cutoff on sampler draws, then measure how much of the original vector distribution falls within it.